Esp32Tap — Assembled-Board Cluster Audit and Test

Guide mode: Assembled-board audit · Isolate the smallest safe boundary, inspect, measure forward, restore exactly, then sign.

Audit the assembled board without wholesale disassembly. No treadmill cable before clusters 1–10; do not connect a treadmill cable until Cluster 11. Never externally drive connected GPIO15 or GPIO21: remove the exact named jumper first. No physical placement or hole coordinate is prescribed.

Device serial: **Hardware revision:**	**Operator:** **Date:**

Source manifest: esp32tap-breadboard-netlist.csv; esp32tap-breadboard-clusters.html; esp32tap-breadboard-from-to.html; DigiKey invoice 129832848; build-guide cluster metadata.

GPIO15 jumper: ____________________ **GPIO21 jumper:** ____________________ **Firmware identity:** ____________________	**Relay exerciser identity:** ____________________ **Observer identity:** ____________________ **Observer manifest:** ____________________

Source contract

Mutually exclusive source rule: USB and STANDALONE POWER are mutually exclusive—one source only. USB connected means STANDALONE POWER physically removed. STANDALONE POWER installed means USB physically disconnected. VBUS sensing is active-low; never join or connect VBUS to a local rail. All sources off before any RJ45 attachment/removal, harness change, or direct-path change. COIL POWER stays open through the unloaded test.

Safety contract

- Initial source: 8.00 V, initial 250 mA current limit; coil-open current below 50 mA. VIN 7.20–7.90 V; LOGIC_3V3 3.20–3.40 V.
- TREAD_OK is low below UV, high at 8.00 V, and low above OV. UV rising 6.25–6.55 V; OV falling 10.30–10.90 V.
- TPS709 enabled 4.75–5.25 V; TPS709 disabled below 0.25 V. Loaded relay limit no more than 500 mA; coil 90–110 mA and at least 4.50 V; BC337 VCE no more than 0.30 V.
- Feedback (1,0) energized and (0,1) bypass; 00 or 11 fault. Remove USB logic power: NC bypass releases within 100 ms. Remove GPIO21 command: NC bypass releases within 100 ms. Remove TREAD_OK: NC bypass releases within 100 ms. Remove VIN: NC bypass releases within 100 ms.
- Five-minute hold: TPS709 and BC337 no more than 45°C and no more than 10°C over ambient. Fifteen-minute bypass hold: every endpoint no more than 40°C and no more than 10°C over ambient. Any unexpected reset is a STOP.

Firmware: The current no-control diagnostic cannot pass a relay exercise gate and cannot pass a UART exercise gate. Coil/local TX tests require an exact bounded relay exerciser identity. Standalone/bypass requires observer identity and manifest proving relay and TX disabled. Cluster 7 is local-only. Cluster 11 is end-to-end: CONSOLE.6 ↔ MOTOR.6. No treadmill relay transfer or MCU TX. A future qualified functional-firmware identity and matching production safety evidence are required by a separate procedure.

Cluster 1 — Raw protection

Dependencies: Empty board and verified bench equipment
Inputs: +8V_RAW, GND

Outputs: VIN, FUSED_8V
Source state: USB absent; STANDALONE POWER removed; COIL POWER removed; bench 8.00 V, initial 250 mA limit

Isolate

Instruction · named link	Operator evidence
Verify STANDALONE POWER and COIL POWER are open. · No link opened — named downstream links verified open	

Inspect

Instruction · orientation	Operator evidence
Inspect F1, D1, D2, and CIN against board marks; D1 cathode stripe faces VIN, D2 cathode faces VIN, and CIN negative faces GND. · D1/D2 cathode stripes toward VIN; CIN negative stripe toward GND; F1 non-polar	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Check protection input-to-output path and absence of a VIN-to-GND short. · F1/D1/D2/CIN · leads and terminals · +8V_RAW, FUSED_8V, VIN, GND · Forward path present; reverse/open behavior and no rail short **device:** F1/D1/D2/CIN · **pin:** leads and terminals · **net:** +8V_RAW, FUSED_8V, VIN, GND · **evidence:** Record continuity direction and VIN–GND resistance	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Apply the smallest safe source and measure forward through F1 and D1. · F1/D1 · input, output, anode, cathode · +8V_RAW → FUSED_8V → VIN · Bench 8.00 V, initial 250 mA limit · STANDALONE POWER removed; COIL POWER removed; GPIO15 and GPIO21 untouched · Meter identity; firmware not required · USB absent; treadmill absent; coil open · VIN 7.20–7.90 V; coil-open current below 50 mA	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify bench polarity, current limit, and GND reference.	
component · Check F1 heating/continuity, D1 direction/drop, D2 leakage, and CIN polarity.	
output · Trace FUSED_8V then VIN; stop at the first missing or current-limited node.	

Restore

Instruction · exact named link	Operator evidence
Remove bench power and leave the verified STANDALONE POWER and COIL POWER links in their documented incoming state. · No link opened	

Dedicated evidence fields: operator · date · signed_pass · bench_voltage_v · input_current_ma · fused_8v_v · vin_v

STOP gate: Current approaches 250 mA, VIN is outside 7.20–7.90 V, polarity is wrong, or anything warms. Remove power; do not build Cluster 2.

PASS gate: Protection chain, polarity, VIN window, and coil-open current pass. PASS — continue to Cluster 2 only. **PASS — continue**

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

Cluster 2 — TSR supply

Dependencies: Cluster 1 **Inputs:** VIN, GND	**Outputs:** TSR_3V3 **Source state:** Cluster 1 source state retained; STANDALONE POWER remains removed

Isolate

Instruction · named link	Operator evidence
With all sources off, open the named STANDALONE POWER link to isolate LOGIC_3V3; leave U2 and its capacitors assembled. · STANDALONE POWER	

Inspect

Instruction · orientation	Operator evidence
Inspect U2 pin order and C2/C3 polarity without removing them; input is VIN, common is GND, output is TSR_3V3. · U2 pins 1/2/3 = VIN/GND/TSR_3V3; C2/C3 negative stripes to GND	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Prove TSR_3V3 is isolated from LOGIC_3V3 and neither rail is shorted to GND. · U2/C2/C3 · pins 1/2/3 and capacitor leads · VIN, GND, TSR_3V3, LOGIC_3V3 · TSR_3V3 not joined to LOGIC_3V3; no rail short **device:** U2/C2/C3 · **pin:** pins 1/2/3 and capacitor leads · **net:** VIN, GND, TSR_3V3, LOGIC_3V3 · **evidence:** Record resistance and link-open continuity	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Apply verified VIN and measure U2 input, common, then output. · U2 · pin 1, pin 2, pin 3 · VIN, GND, TSR_3V3 · Verified Cluster 1 VIN · STANDALONE POWER open; COIL POWER removed; GPIO15 and GPIO21 untouched · Meter identity; firmware not required · USB absent; treadmill absent · TSR_3V3 3.20–3.40 V; LOGIC_3V3 remains unpowered	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify Cluster 1 VIN at U2 pin 1 and GND at pin 2.	
component · Check U2 orientation, C2/C3 polarity, and solder joints.	
output · Measure pin 3 directly, then trace TSR_3V3 to the open STANDALONE POWER boundary.	

Restore

Instruction · exact named link	Operator evidence
Remove power, then restore the exact named STANDALONE POWER link to its recorded incoming state. · STANDALONE POWER	

Dedicated evidence fields: operator · date · signed_pass · vin_v · tsr_3v3_v · logic_3v3_isolation_ohm

STOP gate: TSR_3V3 is outside 3.20–3.40 V, polarity is wrong, LOGIC_3V3 is energized across the removed link, or current/temperature rises.

PASS gate: Verified Cluster 1 VIN produces isolated TSR_3V3 in range. PASS — continue to Cluster 3 only. **PASS — continue**

Operator: ____________ Date: ____________ Signed PASS: ____________

Cluster 3 — DevKit and logic supply

Dependencies: Cluster 2; Verified USB bench source **Inputs:** TSR_3V3, USB_5V, GND	**Outputs:** LOGIC_3V3, DevKit GND, GPIO21_CMD_POST, GPIO15_TX_ENABLE_POST, GPIO17_TX, GPIO38, USB_5V **Source state:** First USB-only with STANDALONE POWER removed; then USB physically disconnected before standalone link installation

Isolate

Instruction · named link	Operator evidence
All sources off: open the named STANDALONE POWER link before USB-only checks; physically disconnect USB before the standalone-only check. · STANDALONE POWER	

Inspect

Instruction · orientation	Operator evidence
Inspect DevKit orientation, antenna clearance, USB connector identity, C4 placement, and both removable command jumper posts. · DevKit USB end and antenna match assembly drawing; C4 spans LOGIC_3V3/GND; command posts labeled	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Confirm grounds map, STANDALONE POWER is open, and USB VBUS is not joined to LOGIC_3V3. · U1/C4/JP1 · 3V3, GND, 5V, command posts, link terminals · LOGIC_3V3, GND, USB_5V, TSR_3V3 · GND common; TSR_3V3 isolated from LOGIC_3V3; VBUS not tied to local 3.3 V **device:** U1/C4/JP1 · **pin:** 3V3, GND, 5V, command posts, link terminals · **net:** LOGIC_3V3, GND, USB_5V, TSR_3V3 · **evidence:** Record mapping and resistance	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Boot USB-only and record the logic rail and boot stability. · U1 · 3V3, GND, USB connector · USB_5V → LOGIC_3V3 · Verified USB bench source · STANDALONE POWER open; command jumpers installed but never externally driven · Exact observer build identity · USB-only; bench VIN absent · LOGIC_3V3 3.20–3.40 V; stable boot; no backfeed	
Disconnect USB, install STANDALONE POWER, and repeat the boot observation. · U1/JP1 · 3V3 and link terminals · TSR_3V3 → LOGIC_3V3 · Verified Cluster 2 TSR_3V3 · USB physically disconnected; STANDALONE POWER installed; GPIO15/GPIO21 not driven · Exact observer build identity · Standalone-only; relay off; TX disabled · LOGIC_3V3 3.20–3.40 V; stable boot; USB_5V unpowered	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify exactly one source is present and the source handoff order was followed.	
component · Check JP1 contact, C4, DevKit seating/orientation, and USB cable/source.	
output · Probe LOGIC_3V3 at U1, then USB_5V and the two command posts for unexpected drive.	

Restore

Instruction · exact named link	Operator evidence
All sources off: restore the exact STANDALONE POWER link to its documented incoming state; reconnect no source during restoration. · STANDALONE POWER	

Dedicated evidence fields: operator · date · signed_pass · usb_logic_3v3_v · standalone_logic_3v3_v · boot_observation

STOP gate: Sources overlap, either source backfeeds the other, LOGIC_3V3 is outside 3.20–3.40 V, or DevKit boot is unstable.

PASS gate: USB-only and standalone-only handoff both pass, with the other source physically absent. PASS — continue to Cluster 4 only. **PASS — continue**

Operator: ____________ Date: ____________ Signed PASS: ____________

Cluster 4 — TPS3700 voltage monitor

Dependencies: Cluster 1; Cluster 3 **Inputs:** VIN, LOGIC_3V3, GND	**Outputs:** UV_SENSE, OV_SENSE, TREAD_OK, TREAD_OK_MCU **Source state:** USB logic only; STANDALONE POWER removed; bench source swept at protected VIN

Isolate

Instruction · named link	Operator evidence
Keep command jumpers connected for observation only; open no assembled divider link and sweep only protected VIN with USB logic supplied. · No link opened — GPIO6 observation only	

Inspect

Instruction · orientation	Operator evidence
Inspect U3 notch/pin 1, divider resistor identities, adapter mapping, and pull-up orientation before power. · U3 notch and pin 1 mark verified; UV and OV divider populations match labels; adapter map recorded	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Measure each divider leg and map adapter pins to U3 and GPIO6. · U3/R-divider/adapter · U3 sense/output pins and adapter pins · VIN, UV_SENSE, OV_SENSE, TREAD_OK, GPIO6 · UV path implements VIN / 16; OV path implements VIN / 26.5; output reaches GPIO6 **device:** U3/R-divider/adapter · **pin:** U3 sense/output pins and adapter pins · **net:** VIN, UV_SENSE, OV_SENSE, TREAD_OK, GPIO6 · **evidence:** Record resistance ratios and continuity map	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Sweep protected VIN upward and downward while observing both sense pins and TREAD_OK. · U3 · sense pins, output pin · VIN, UV_SENSE, OV_SENSE, TREAD_OK · Protected VIN sweep below UV → 8.00 V → above OV → downward · STANDALONE POWER removed; GPIO15/GPIO21 untouched · Exact observer build identity · USB logic only; GPIO6 is high-impedance observation · TREAD_OK low below UV, high at 8.00 V, low above OV; UV rising 6.25–6.55 V; OV falling 10.30–10.90 V	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify protected VIN sweep, USB logic rail, and common GND.	
component · Check divider values, U3 orientation, pull-up, and adapter mapping.	
output · Probe UV_SENSE and OV_SENSE before TREAD_OK, then GPIO6.	

Restore

Instruction · exact named link	Operator evidence
Return VIN to zero, remove all power, and leave the unchanged observation wiring in its recorded incoming state. · No link opened	

Dedicated evidence fields: operator · date · signed_pass · uv_rising_v · ov_falling_v · tread_ok_low_v · tread_ok_high_v

STOP gate: TREAD_OK three-state behavior or thresholds fail; never exceed protected VIN about 11.1 V; stop if TVS draws appreciable current or warms.

PASS gate: Both sweep directions and GPIO6 isolated observation match limits. PASS — continue to Cluster 5 only. **PASS — continue**

Operator: ____________ Date: ____________ Signed PASS: ____________

Cluster 5 — AHC08 permission logic

Dependencies: Cluster 3; Cluster 4
Inputs: LOGIC_3V3, TREAD_OK, RELAY_CMD, TX_ENABLE, GND

Outputs: RELAY_GATE, TX_GATE
Source state: USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

Isolate

Instruction · named link	Operator evidence
All sources off: remove the exact GPIO21 jumper before any RELAY_CMD manual injection; remove the exact GPIO15 jumper before any TX_ENABLE manual injection. · GPIO21 jumper	
Keep sources off and remove the second named command boundary before its manual truth-table injection. · GPIO15 jumper	

Inspect

Instruction · orientation	Operator evidence
Inspect U4 notch/pin 1, decoupling, pull-downs, and labeled jumper endpoints; never externally drive a connected DevKit GPIO. · U4 notch and pin 1 verified; GPIO21 ↔ RELAY_CMD and GPIO15 ↔ TX_ENABLE endpoints labeled	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Map permission-logic inputs and outputs with both command jumpers removed. · U4 · input and output pins · RELAY_CMD, TX_ENABLE, TREAD_OK, RELAY_GATE, TX_GATE · Exact map; command inputs default low; no output short **device:** U4 · **pin:** input and output pins · **net:** RELAY_CMD, TX_ENABLE, TREAD_OK, RELAY_GATE, TX_GATE · **evidence:** Record pin/net map and pull-down resistance	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Inject only at the isolated RELAY_CMD side and execute the four-state gate table. · U4 · RELAY_CMD input, TREAD_OK input, RELAY_GATE output · RELAY_CMD + TREAD_OK → RELAY_GATE · Current-limited 0 V/3.3 V manual injection · GPIO21 jumper removed exactly; GPIO15 jumper remains removed during paired audit · Meter/scope identity; firmware not used to source the input · DevKit disconnected from injected node · RELAY_GATE high only when both inputs are high	
Inject only at the isolated TX_ENABLE side and execute the four-state gate table. · U4 · TX_ENABLE input, TREAD_OK input, TX_GATE output · TX_ENABLE + TREAD_OK → TX_GATE · Current-limited 0 V/3.3 V manual injection · GPIO15 jumper removed exactly; GPIO21 jumper remains removed during paired audit · Meter/scope identity; firmware not used to source the input · DevKit disconnected from injected node · TX_GATE high only when both inputs are high	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify LOGIC_3V3, GND, TREAD_OK state, and that the exact command jumper is removed.	
component · Check U4 orientation, decoupling, pull-downs, and pin map.	
output · Probe each isolated input at U4, then its gate output; stop at first mismatch.	

Restore

Instruction · exact named link	Operator evidence
All injection and sources off: restore the exact GPIO21 jumper and verify RELAY_CMD is no longer externally driven. · GPIO21 jumper	
With injection absent, restore the exact GPIO15 jumper and verify TX_ENABLE is no longer externally driven. · GPIO15 jumper	

Dedicated evidence fields: operator · date · signed_pass · relay_truth_table · tx_truth_table · gpio21_jumper_restore · gpio15_jumper_restore

STOP gate: Any gate rises without both inputs, a command GPIO remains connected during manual drive, or supply/reset behavior is abnormal.

PASS gate: Both four-state truth tables pass and temporary drives are removed. PASS — continue to Cluster 6 only. **PASS — continue**

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

Cluster 6 — TPS709 and BC337 driver

Dependencies: Cluster 1; Cluster 5 **Inputs:** VIN, RELAY_GATE, GND	**Outputs:** +5V_RLY, RELAY_COIL− **Source state:** Valid VIN/TREAD_OK; COIL POWER remains removed for the unloaded test

Isolate

Instruction · named link	Operator evidence
Verify the named COIL POWER link is open; do not disturb the assembled regulator or BC337 wiring. · COIL POWER — verified open, not newly opened	
All sources off: remove the exact GPIO21 jumper before bounded RELAY_CMD injection. · GPIO21 jumper	

Inspect

Instruction · orientation	Operator evidence
Inspect U5 orientation, Q1 BC337 C/B/E mapping, D3 polarity, and C5/C6 polarity before the unloaded audit. · U5 pin 1 mark verified; BC337 collector/base/emitter physically mapped; D3 cathode toward +5V_RLY	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Map regulator enable/output, base resistor, BC337 pins, diode, and the open coil boundary. · U5/Q1/D3 · enable/output and C/B/E pins · RELAY_GATE, +5V_RLY, COIL_LOW, GND · Exact device-pin/net map; coil remains isolated; no +5V_RLY short **device:** U5/Q1/D3 · **pin:** enable/output and C/B/E pins · **net:** RELAY_GATE, +5V_RLY, COIL_LOW, GND · **evidence:** Record continuity, transistor map, and COIL POWER isolation	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Measure disabled then enabled regulator output and base/collector behavior with the coil isolated. · U4/U5/Q1 · RELAY_CMD input, RELAY_GATE output, regulator enable/output, BC337 base/collector/emitter · RELAY_CMD → RELAY_GATE → +5V_RLY and BC337 base drive · Bounded RELAY_CMD injection, 0/3.3 V through 1 kΩ, with valid TREAD_OK high · COIL POWER open; GPIO21 jumper removed before bounded RELAY_CMD injection · Bench injector identity; output current limited by 1 kΩ · AHC08 remains powered at valid TREAD_OK; no relay coil load; treadmill absent · TPS709 enabled 4.75–5.25 V; TPS709 disabled below 0.25 V; output discharges	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify VIN, LOGIC_3V3, GND, bounded RELAY_CMD stimulus, valid TREAD_OK, and COIL POWER open.	
component · Check U5 orientation/thermal pad, Q1 C/B/E map, D3 polarity, resistors, and capacitors.	
output · Probe U5 enable then +5V_RLY, Q1 base, collector, and emitter in order.	

Restore

Instruction · exact named link	Operator evidence
Remove bounded RELAY_CMD injection and all power; leave COIL POWER in the documented incoming open state. · COIL POWER — remains open	
With injection absent and all sources off, restore the exact GPIO21 jumper. · GPIO21 jumper	

Dedicated evidence fields: operator · date · signed_pass · tps709_disabled_v · tps709_enabled_v · bc337_base_v · coil_power_open

STOP gate: COIL POWER is installed, enabled or disabled limits fail, output does not discharge below 0.25 V, or Q1/U5 heats.

PASS gate: Unloaded regulator and base drive pass with coil isolated. PASS — continue to Cluster 7 only. **PASS — continue**

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

Cluster 7 — Relay coil, local contacts, and feedback

Dependencies: Cluster 6; Cluster 3
Inputs: +5V_RLY, RELAY_COIL−, LOGIC_3V3, GND

Outputs: LOCAL_NC, LOCAL_NO, FB_NC, FB_NO
Source state: Local-only bench exercise; treadmill/RJ45 cables absent; bounded relay exerciser verified; loaded relay limit no more than 500 mA

Isolate

Instruction · named link	Operator evidence
Disconnect treadmill/RJ45 cables; verify GPIO21 jumper remains installed so bounded relay exerciser firmware is the sole RELAY_CMD source. · GPIO21 jumper remains installed	

Inspect

Instruction · orientation	Operator evidence
Inspect K1 orientation, coil diode polarity, local NC/NO/COM labels, feedback contact mapping, and coil terminals. · K1 pin map and contact symbols verified; flyback diode cathode toward +5V_RLY; local endpoints labeled	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Map the coil, local contact sets, and feedback contacts with COIL POWER open. · K1/D3/Q1 · coil pins, COM/NC/NO pins, feedback pins · +5V_RLY, COIL_LOW, LOCAL_NC, LOCAL_NO, RELAY_FB_NC, RELAY_FB_NO · Coil finite and not shorted; local COM→NC continuity; feedback reports bypass **device:** K1/D3/Q1 · **pin:** coil pins, COM/NC/NO pins, feedback pins · **net:** +5V_RLY, COIL_LOW, LOCAL_NC, LOCAL_NO, RELAY_FB_NC, RELAY_FB_NO · **evidence:** Record coil resistance and each de-energized contact map	

Step / state ledger — re-arm before every destructive release

Stage · required state · action · evidence	Operator record
rearm_usb_loss · USB logic powered; VIN 8.00 V; TREAD_OK high; GPIO21 asserted by bounded firmware · Install COIL POWER under limit; boot exact bounded relay exerciser firmware and command energized state · Confirm relay energized; feedback (1,0); local NO selected	
release_usb · Re-arm evidence signed; passive scope armed; no external command fixture · Disconnect USB logic power; do not apply external drive · Record NC return and USB logic release within 100 ms	
rearm_gpio21 · USB logic powered; VIN 8.00 V; TREAD_OK high; GPIO21 asserted by bounded firmware · Restore USB logic power; boot exact firmware; command energized state again · Confirm relay energized; feedback (1,0); local NO selected	
release_gpio21 · Re-arm evidence signed; USB logic remains powered; passive scope armed · Have firmware deassert GPIO21; do not remove power or inject externally · Record NC return and GPIO21 release within 100 ms	
rearm_tread_ok · USB logic powered; VIN 8.00 V; TREAD_OK high; GPIO21 asserted by bounded firmware · Return VIN to 8.00 V; verify TREAD_OK high; command energized state again · Confirm relay energized; feedback (1,0); local NO selected	
release_tread_ok · Re-arm evidence signed; USB logic remains powered; passive scope armed · Lower protected VIN below UV to remove TREAD_OK; never drive an unpowered AHC08 · Record NC return and TREAD_OK release within 100 ms	
rearm_vin · USB logic powered; VIN 8.00 V; TREAD_OK high; GPIO21 asserted by bounded firmware · Return VIN to 8.00 V; verify TREAD_OK high; command energized state again · Confirm relay energized; feedback (1,0); local NO selected	
release_vin · Re-arm evidence signed; USB logic remains powered; passive scope armed · Switch bench VIN off; leave GPIO21 under powered firmware control · Record NC return and VIN release within 100 ms	
rearm_thermal · USB logic powered; VIN 8.00 V; TREAD_OK high; GPIO21 asserted by bounded firmware · Restore USB, VIN, TREAD_OK, firmware command, and loaded current limit · Confirm relay energized; feedback (1,0); local NO selected	
thermal_hold · Re-arm evidence signed; ambient recorded; no treadmill/RJ45 cable · Run bounded five-minute firmware hold, then command relay off · Record ambient, TPS709, BC337, coil, feedback, and final NC return	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Install COIL POWER under the loaded limit and command one bounded local energization. · U1/U4/K1/U5/Q1 · GPIO21, RELAY_CMD, RELAY_GATE, coil terminals, Q1 collector/emitter, feedback contacts · GPIO21 → RELAY_CMD → RELAY_GATE → coil and local contacts · Bounded relay exerciser firmware asserts GPIO21 for one timed pulse then deasserts · GPIO21 jumper remains installed; GPIO15 untouched; COIL POWER installed under limit · Exact bounded relay exerciser firmware identity and SHA · Local-only; treadmill/RJ45 absent; loaded limit no more than 500 mA · Coil 90–110 mA; coil voltage at least 4.50 V; BC337 VCE no more than 0.30 V; feedback (1,0) energized and (0,1) bypass; 00 or 11 fault	
Measure release after removing USB logic power. · U1/U4/K1 · GPIO21, local COM/NC, feedback pins · USB logic removal → GPIO21 drive loss → NC bypass · Bounded relay exerciser firmware asserts GPIO21; disconnect USB logic power while energized · GPIO21 jumper remains installed; COIL POWER installed only for bounded test · Exact bounded relay exerciser firmware identity and SHA · USB disconnect removes GPIO21 drive and AHC08 logic power; passive scope on local contact; treadmill absent · USB logic release within 100 ms	
Measure release after removing GPIO21 command. · U1/U4/K1 · GPIO21, local COM/NC, feedback pins · GPIO21 command removal → NC bypass · Bounded relay exerciser firmware deasserts GPIO21 while USB logic remains powered · GPIO21 jumper remains installed; COIL POWER installed · Exact bounded relay exerciser firmware identity and SHA · USB logic powered; AHC08 active; passive scope on local contact; treadmill absent · GPIO21 release within 100 ms	

Cluster 7 — Relay coil, local contacts, and feedback (continued)

Measure — powered release / thermal evidence (continued)

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Measure release after removing TREAD_OK. · U1/U3/U4/K1 · GPIO21, TREAD_OK, local COM/NC, feedback pins · TREAD_OK removal → NC bypass · Firmware holds GPIO21 high; lower protected bench VIN below UV to deassert TREAD_OK · GPIO21 jumper remains installed; COIL POWER installed · Exact bounded relay exerciser firmware identity and SHA · USB logic remains powered; AHC08 observes TREAD_OK low; passive scope on local contact · TREAD_OK release within 100 ms	________________
Measure release after removing VIN. · U1/U3/U4/U5/K1 · GPIO21, VIN, local COM/NC, feedback pins · VIN removal → NC bypass · Firmware holds GPIO21 high; switch bench VIN off while USB logic remains powered · GPIO21 jumper remains installed; COIL POWER installed · Exact bounded relay exerciser firmware identity and SHA · USB logic remains powered; AHC08 is never externally driven unpowered; passive scope on local contact · VIN release within 100 ms	________________
Run the five-minute local thermal hold and record all temperatures. · U1/U4/U5/Q1/K1 · GPIO21, regulator case, BC337 case, coil terminals · Sustained bounded firmware GPIO21 command · Bounded relay exerciser firmware five-minute hold under no more than 500 mA source limit · GPIO21 jumper remains installed; COIL POWER installed; treadmill absent · Exact bounded relay exerciser firmware identity and SHA · Ambient, TPS709, BC337, and relay observed · TPS709 and BC337 no more than 45°C and no more than 10°C over ambient	________________

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify loaded current limit, valid VIN/TREAD_OK, logic power, and bounded exerciser identity.	________________
component · Check K1 coil/contact pin map, U5 output, Q1 C/B/E, diode, and feedback wiring.	________________
output · Measure +5V_RLY, coil voltage/current, local contact transfer, then feedback outputs.	________________

Restore

Instruction · exact named link	Operator evidence
All power off: remove COIL POWER if it was installed; leave the GPIO21 jumper installed in its documented incoming state. · GPIO21 jumper remains installed	________________

Dedicated evidence fields: operator · date · signed_pass · relay_exerciser_identity · usb_logic_release_ms · gpio21_release_ms · tread_ok_release_ms · vin_release_ms · ambient_temp_c · tps709_temp_c · bc337_temp_c

Dedicated release / thermal evidence: Relay exerciser identity ______ · USB logic release (ms) ______ · GPIO21 release (ms) ______ · TREAD_OK release (ms) ______ · VIN release (ms) ______ · Ambient temperature (°C) ______ · TPS709 temperature (°C) ______ · BC337 temperature (°C) ______

STOP gate: Any electrical, feedback, release, reset, or five-minute thermal gate fails. Cluster 7 is local-only; it makes no end-to-end RJ45 claim.

PASS gate: Local contacts, feedback, current, voltage, release, and thermal evidence all pass. PASS — continue to Cluster 8 only. **PASS — continue**

Operator: ____________________ Date: ____________________ Signed PASS: ____________________

Cluster 8 — AHC126 and UART taps

Dependencies: Cluster 3; Cluster 5; Cluster 7
Inputs: LOGIC_3V3, TX_GATE, GPIO17_TX, LOCAL_NC, LOCAL_NO, GND

Outputs: TX_DRV, UART_BUFFER_HIGH_Z, UART_BUFFER_FOLLOWS, GPIO18_RX_LOCAL, GPIO16_RX_LOCAL, LOCAL_TX_SELECTED
Source state: Local future-interface fixture only; treadmill and both RJ45 breakouts absent; bounded UART/relay exerciser identity recorded

Isolate

Instruction · named link	Operator evidence
Disconnect both RJ45 breakouts and remove the exact GPIO15 jumper before manual TX_ENABLE drive; GPIO21 jumper remains installed for bounded relay exerciser firmware. · GPIO15 jumper	

Inspect

Instruction · orientation	Operator evidence
Inspect U6 notch/pin 1, output-enable polarity, 10 kΩ receive-tap resistors, and local stub/contact labels. · U6 notch and pin 1 verified; receive resistors are 10 kΩ; local NC/NO destinations identified	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Map receive-tap resistors and the relay-selected local TX path with connector sides absent. · U6/R-taps/K1 · AHC126 channels, resistor ends, local relay contacts · LOCAL_CONSOLE_6_STUB, LOCAL_PIN3_STUB, GPIO18_RX_LOCAL, GPIO16_RX_LOCAL, LOCAL_TX_SELECTED · Each receive tap passes through 10 kΩ and stays isolated; TX selection maps only local contacts **device:** U6/R-taps/K1 · **pin:** AHC126 channels, resistor ends, local relay contacts · **net:** LOCAL_CONSOLE_6_STUB, LOCAL_PIN3_STUB, GPIO18_RX_LOCAL, GPIO16_RX_LOCAL, LOCAL_TX_SELECTED · **evidence:** Record isolation, resistance, and local contact map	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Drive LOCAL_CONSOLE_6_STUB low/high and observe only GPIO18_RX_LOCAL. · U6/R-tap · receive input/output channel and GPIO18 · LOCAL_CONSOLE_6_STUB → GPIO18_RX_LOCAL · Bounded UART exerciser low/high pattern · GPIO15 jumper removed; GPIO21 jumper remains installed for firmware relay control · Exact bounded UART and relay exerciser firmware identities · Treadmill and RJ45 absent; local receive only · GPIO18_RX_LOCAL follows valid low/high without loading source	
Drive LOCAL_PIN3_STUB low/high and observe only GPIO16_RX_LOCAL. · U6/R-tap · receive input/output channel and GPIO16 · LOCAL_PIN3_STUB → GPIO16_RX_LOCAL · Bounded UART exerciser low/high pattern · GPIO15 jumper removed; GPIO21 jumper remains installed for firmware relay control · Exact bounded UART and relay exerciser firmware identities · Treadmill and RJ45 absent; local receive only · GPIO16_RX_LOCAL follows valid low/high without loading source	
Prove TX_DRV high impedance while disabled, then bounded following and relay-selected local TX on each local contact state. · U1/U4/U6/K1 · GPIO21, TX input, enable, TX_DRV, local NC/NO contacts · TX_GATE + GPIO17_TX → TX_DRV → LOCAL_TX_SELECTED · Bounded UART waveform plus bounded relay exerciser firmware toggling GPIO21 · GPIO15 jumper removed before manual TX_ENABLE drive; GPIO21 jumper remains installed for firmware relay transfer · Exact bounded UART and relay exerciser firmware identities · Local-only TX load; no connector-side wire; firmware is sole relay command source · TX_DRV high impedance disabled; enabled waveform follows; LOCAL_TX_SELECTED matches NC then NO selection	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify LOGIC_3V3, TREAD_OK, bounded exerciser identity, and both command-jumper states.	
component · Check U6 orientation/OE polarity, 10 kΩ taps, and K1 local contact map.	
output · Probe local input stub, receive GPIO, TX_DRV, then LOCAL_TX_SELECTED in signal order.	

Restore

Instruction · exact named link	Operator evidence
Remove manual TX drive and all power, then restore the exact GPIO15 jumper; leave GPIO21 installed in its unchanged incoming state. · GPIO15 jumper	

Dedicated evidence fields: operator · date · signed_pass · relay_exerciser_identity · gpio18_rx_local_low_v · gpio18_rx_local_high_v · gpio16_rx_local_low_v · gpio16_rx_local_high_v · local_tx_selected_nc_ohm · local_tx_selected_no_ohm

Dedicated local interface evidence: GPIO18_RX_LOCAL low (V) ______ high (V) ______ · GPIO16_RX_LOCAL low (V) ______ high (V) ______ · LOCAL_TX_SELECTED NC (Ω) ______ NO (Ω) ______ · Relay exerciser identity ______

STOP gate: TX_DRV drives while disabled, enabled waveform does not follow, either receive tap fails, local relay selection fails, any connector-side wire is installed, or exerciser identity is absent.

PASS gate: Local high impedance, following, receive taps, and relay-selected TX pass; connector-side checks remain deferred. PASS — continue to Cluster 9 only.
PASS — continue

Operator: ____________ Date: ____________ Signed PASS: ____________

Cluster 9 — Indicators and VBUS sensing

Dependencies: Cluster 3
Inputs: LOGIC_3V3, GPIO38, USB_5V, GND

Outputs: POWER_LED, STATUS_LED, VBUS_PRESENT_N
Source state: USB-only test then standalone-only test; never join VBUS to a local rail

Isolate

Instruction · named link	Operator evidence
Open the named STANDALONE POWER link for USB-only observation; never join VBUS to any local rail. · STANDALONE POWER	

Inspect

Instruction · orientation	Operator evidence
Inspect Q2 orientation, D4/D5 polarity, LED resistors, C7, and the VBUS active-low sense divider. · Q2 G/D/S mapped; LED cathodes face GND; C7 spans LOGIC_3V3/GND; VBUS path remains isolated	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Map LED branches and prove VBUS is isolated from VIN, +5V_RLY, and LOGIC_3V3. · Q2/D4/D5/C7 · transistor pins, LED leads, resistor ends, capacitor leads · USB_5V, USB_SENSE_N, LOGIC_3V3, GPIO38, GND · LED polarity correct; VBUS has no power-rail continuity; active-low sense path only **device:** Q2/D4/D5/C7 · **pin:** transistor pins, LED leads, resistor ends, capacitor leads · **net:** USB_5V, USB_SENSE_N, LOGIC_3V3, GPIO38, GND · **evidence:** Record polarity, mapping, and cross-rail resistance	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Observe power and status LEDs plus the active-low USB presence input under USB-only then standalone-only sources. · Q2/D4/D5/U1 · LED nodes, USB sense node, GPIO input · USB_5V present/absent and GPIO38 state · USB-only, then physically disconnected before standalone-only · STANDALONE POWER open for USB-only; installed only after USB removal; command jumpers untouched · Exact observer build identity · One source only; relay off; TX disabled · LOGIC_3V3 3.20–3.40 V; LED polarity/current limiting correct; VBUS sense active-low; no VBUS backfeed	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify exclusive source state, USB cable presence, and LOGIC_3V3.	
component · Check Q2 G/D/S, LED polarity/resistors, C7, and sense-divider parts.	
output · Probe USB_5V, USB_SENSE_N, GPIO input, then each LED node.	

Restore

Instruction · exact named link	Operator evidence
All sources off: restore the exact STANDALONE POWER link to its documented incoming state and leave USB physically disconnected if installed. · STANDALONE POWER	

Dedicated evidence fields: operator · date · signed_pass · power_led_observation · status_led_observation · vbus_present_logic · vbus_absent_logic

STOP gate: VBUS joins a local rail, active-low truth is wrong, LED polarity/current limiting is wrong, or sources overlap.

PASS gate: Both LEDs and isolated active-low VBUS sense pass under exclusive sources. PASS — continue to Cluster 10 only. **PASS — continue**

Operator: ______________ Date: ______________ Signed PASS: ______________

Cluster 10 — Whole-device standalone bench test

Dependencies: Cluster 1; Cluster 2; Cluster 3; Cluster 4; Cluster 5; Cluster 6; Cluster 7; Cluster 8; Cluster 9
Inputs: +8V_RAW, GND, observer firmware

Outputs: standalone boot, Wi-Fi observation, event log, UART idle evidence, verified observer state
Source state: USB physically absent; STANDALONE POWER installed; observer verified; relay off; TX disabled

Isolate

Instruction · named link	Operator evidence
Physically disconnect USB; install STANDALONE POWER only after USB is absent; leave COIL POWER in its documented state and attach no treadmill cable. · No link opened — exclusive source boundary established	

Inspect

Instruction · orientation	Operator evidence
Inspect source-link labels, observer manifest identity, antenna clearance, and UART probe points before standalone boot. · USB absent; STANDALONE POWER identity verified; GPIO16/GPIO18 probe points labeled; relay/TX disabled manifest available	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Confirm USB VBUS is absent and the standalone path maps TSR_3V3 to LOGIC_3V3 without shorts. · U1/JP1/UART taps · USB, link terminals, 3V3/GND, GPIO16/GPIO18 · USB_5V, TSR_3V3, LOGIC_3V3, GND · USB physically absent; standalone link map correct; no rail short **device:** U1/JP1/UART taps · **pin:** USB, link terminals, 3V3/GND, GPIO16/GPIO18 · **net:** USB_5V, TSR_3V3, LOGIC_3V3, GND · **evidence:** Record source isolation and rail resistance	

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Boot the identified observer standalone and record rails, Wi-Fi/event-log observation, relay state, and feedback. · U1/U2/U4/K1 · 3V3, GPIO controls, feedback inputs · TSR_3V3 → LOGIC_3V3 → passive observer · Standalone bench VIN with observer boot · USB absent; STANDALONE POWER installed; relay off; TX disabled · Exact observer identity · Observer manifest proves relay and TX disabled · LOGIC_3V3 3.20–3.40 V; Wi-Fi/event log present; no reset; feedback (0,1) bypass	
Measure GPIO16 UART idle without transmitting. · U1 · GPIO16 · GPIO16_RX_LOCAL idle · Passive high-impedance meter/scope only · USB absent; STANDALONE POWER installed; no jumper removed or external drive · Exact observer identity · Observer manifest; relay off; TX disabled · Record stable GPIO16 UART idle level; no transmitted waveform	
Measure GPIO18 UART idle without transmitting. · U1 · GPIO18 · GPIO18_RX_LOCAL idle · Passive high-impedance meter/scope only · USB absent; STANDALONE POWER installed; no jumper removed or external drive · Exact observer identity · Observer manifest; relay off; TX disabled · Record stable GPIO18 UART idle level; no transmitted waveform	

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify standalone VIN, source exclusivity, observer identity, and manifest.	
component · Check U2/JP1/U1 seating, reset cause, antenna, and passive probe loading.	
output · Measure TSR_3V3 then LOGIC_3V3, boot/event log, GPIO16/GPIO18 idle, and feedback.	

Restore

Instruction · exact named link	Operator evidence
Power off and return STANDALONE POWER to its documented incoming state; leave USB and treadmill cables disconnected. · No link opened	

Dedicated evidence fields: operator · date · signed_pass · observer_identity · observer_manifest · logic_3v3_v · gpio16_idle_v · gpio18_idle_v · wifi_event_log · feedback_state

Dedicated standalone evidence: Observer identity ______ · Observer manifest ______ · GPIO16 UART idle (V) ______ · GPIO18 UART idle (V) ______

STOP gate: Identity/manifest missing, rail out of range, reset, Wi-Fi/event-log failure, UART idle missing, relay enables, TX enables, or feedback differs from (0,1).

PASS gate: Identified observer boots standalone and all passive observations pass. PASS — continue to Cluster 11 only. **PASS — continue**

Operator: ______________________ Date: ______________________ Signed PASS: ______________________

Cluster 11 — RJ45 pass-through and treadmill bypass

Dependencies: Cluster 7; Cluster 8; Cluster 10
Inputs: CONSOLE.1–8, MOTOR.1–8, verified observer state

Outputs: eight mapped conductors, end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass, bypass-only treadmill evidence
Source state: USB absent; standalone installed; observer verified; relay off; TX disabled; source → state → harness → direct-path sequence

Isolate

Instruction · named link	Operator evidence
All sources off: remove the fused-DMM harness and open the first GND common link before individual mapping. · CONSOLE.1 ↔ MOTOR.1 GND link	
Keep all sources off and open the second GND common link before individual mapping. · CONSOLE.7 ↔ MOTOR.7 GND link	
Keep all sources off and open the first +8V_RAW common link before individual mapping. · CONSOLE.2 ↔ MOTOR.2 +8V_RAW link	
Keep all sources off and open the second +8V_RAW common link before individual mapping. · CONSOLE.8 ↔ MOTOR.8 +8V_RAW link	

Inspect

Instruction · orientation	Operator evidence
Inspect CONSOLE/MOTOR identities, pin numbering, dual-pin fused-DMM harness fuses/polarity, strain relief, and every terminal before attachment. · Both connector orientations and pins 1–8 independently identified; harness review identity recorded; no heat/discoloration	

Measure — unpowered evidence

Instruction · points · expected	Recorded evidence
Map all eight conductors independently before any common links are restored. · CONSOLE/MOTOR/K1 · pins 1–8 and local relay contacts · all eight connector paths including CONSOLE.6 ↔ MOTOR.6 · Eight exact maps pass; unrelated pins open during isolated mapping; CONSOLE.6 ↔ MOTOR.6 is NC bypass **device:** CONSOLE/MOTOR/K1 · **pin:** pins 1–8 and local relay contacts · **net:** all eight connector paths including CONSOLE.6 ↔ MOTOR.6 · **evidence:** Record each end-to-end map and unrelated-pin isolation	

Step / state ledger — ground return, harness, and +8V restoration

Stage · required state · action · evidence	Operator record
map_complete · All sources off; four common links open; fused-DMM harness absent · Complete eight independent maps before restoring any common link · Signed map; unrelated pins open; CONSOLE.6 ↔ MOTOR.6 NC path proven	
restore_ground_1 · All sources off; mapping signed; both +8V_RAW links remain open · All sources off after mapping: restore the first exact GND common link. · CONSOLE.1 ↔ MOTOR.1 GND link; signed restoration	
restore_ground_7 · All sources off; first GND link restored; both +8V_RAW links remain open · All sources off after mapping: restore the second exact GND common link. · CONSOLE.7 ↔ MOTOR.7 GND link; signed restoration	
verify_ground_return · All sources off; both GND links restored; both +8V_RAW links open; harness absent · Verify end-to-end ground return continuity and unrelated-pin isolation · Ground return resistance ______ Ω; continuity PASS signed	
install_harness · All sources off; GND links restored; +8V_RAW direct links open; treadmill power remains off · Install reviewed fused DMM harness across +8V_RAW paths only; verify fuses, polarity, and strain relief · Harness identity and installation PASS signed before power	
apply_treadmill_power · Observer verified; USB absent; standalone installed; relay off; TX disabled · Apply treadmill power for bypass-only current measurement · Current no more than 500 mA; no reset, transfer, or TX	
power_off · Current evidence signed; treadmill power present only through reviewed harness · Remove treadmill power and confirm all sources off before changing harness · Zero volts confirmed at both +8V_RAW paths	
remove_harness · All sources off; both GND links remain restored; +8V_RAW direct links remain open · Remove fused DMM harness; inspect both fused leads and terminals · Harness removed with no heat or damage	
restore_plus8_2 · All sources off; harness removed; GND return verified · All sources off before removal of harness; after harness removal, restore the first exact +8V_RAW common link. · CONSOLE.2 ↔ MOTOR.2 +8V_RAW link; signed restoration	
restore_plus8_8 · All sources off; harness removed; first +8V_RAW link restored · All sources off after removal of harness: restore the second exact +8V_RAW common link. · CONSOLE.8 ↔ MOTOR.8 +8V_RAW link; both direct paths verified	

Cluster 11 — RJ45 pass-through and treadmill bypass (powered evidence)

Measure — powered / observed

Instruction · device · pin · net · stimulus · jumper state · firmware identity · observation · limits	Recorded evidence
Perform receive-only mapping observation with no transmit and no relay transfer. · U1/U6 · GPIO16, GPIO18, RX tap pins · CONSOLE receive paths → GPIO16/GPIO18 · Identified observer; treadmill initially unpowered · USB absent; STANDALONE POWER installed; relay off; TX disabled · Exact observer identity · Observer manifest proves relay and TX disabled · Receive-only activity observed; no MCU TX and no treadmill relay transfer	________
Run bypass-only current measurement through the reviewed dual-pin fused-DMM harness. · Fused-DMM harness/CONSOLE/MOTOR · both independent +8V pins and GND return · source → state → harness → direct-path sequence · Apply treadmill power for bypass-only operation after signed harness installation · USB absent; STANDALONE POWER installed; observer verified; relay off; TX disabled · Exact observer identity · Observer manifest and harness review identity recorded · Treadmill current separately measured no more than 500 mA; current no-control diagnostic cannot pass relay exercise gate and cannot pass UART exercise gate	________
After power-off, harness removal, and both +8V_RAW restorations, measure supply and ground-return drops. · CONSOLE/MOTOR · restored supply and ground endpoints · restored direct paths → voltage drop · Bypass-only treadmill after direct-path restoration · USB absent; STANDALONE POWER installed; relay off; TX disabled · Exact observer identity · Observer manifest; GND links restored before harness; +8V links restored after harness removal · Supply drop no more than 50 mV; ground-return drop no more than 50 mV	________
After direct-path restoration, run the fifteen-minute bypass thermal test at every terminal. · CONSOLE/MOTOR/power endpoints · CONSOLE pins 1–8, MOTOR pins 1–8, +8V and GND endpoints · restored direct paths → terminal temperatures · Fifteen-minute bypass treadmill hold · USB absent; STANDALONE POWER installed; relay off; TX disabled · Exact observer identity · Observer manifest; no transfer and no TX · Every terminal and power-path endpoint no more than 40°C and no more than 10°C over ambient	________

Cluster 11 — RJ45 pass-through and treadmill bypass (continued)

Likely causes — ordered source → component → output

Stage · diagnostic	Evidence
source · Verify bypass-only source sequence, observer/manifest, harness identity, fuses, and all-sources-off transition records.	
component · Check connector orientation/terminals, independent +8V conductors, GND returns, relay NC contact, and tap resistors.	
output · Trace source → component → output: each pin map, CONSOLE.6 through local NC to MOTOR.6, current harness, restored drops, then thermal endpoints.	

Restore

Instruction · exact named link	Operator evidence
All sources off before removal of harness; after harness removal, restore the first exact +8V_RAW common link. · CONSOLE.2 ↔ MOTOR.2 +8V_RAW link	
All sources off after removal of harness: restore the second exact +8V_RAW common link. · CONSOLE.8 ↔ MOTOR.8 +8V_RAW link	

Dedicated evidence fields: operator · date · signed_pass · observer_identity · observer_manifest · harness_identity · treadmill_current_ma · supply_drop_mv · ground_return_drop_mv · plus_8v_endpoint_temp_c · gnd_endpoint_temp_c · console_1_temp_c · console_2_temp_c · console_3_temp_c · console_4_temp_c · console_5_temp_c · console_6_temp_c · console_7_temp_c · console_8_temp_c · motor_1_temp_c · motor_2_temp_c · motor_3_temp_c · motor_4_temp_c · motor_5_temp_c · motor_6_temp_c · motor_7_temp_c · motor_8_temp_c

Dedicated bypass and thermal evidence: Observer identity ______ · Observer manifest ______ · Treadmill current (mA) ______ · Supply drop (mV) ______ · Ground-return drop (mV) ______ · CONSOLE temperature (°C), pins 1–8 ____________________ · MOTOR temperature (°C), pins 1–8 ____________________ · +8V endpoint temperature (°C) ______ · GND endpoint temperature (°C) ______

Dedicated C11 record	Writable evidence
Eight independent pin maps and unrelated-pin isolation	
Treadmill current; supply and ground-return drops	
CONSOLE pins 1–8 temperatures	
MOTOR pins 1–8; +8V and GND endpoint temperatures	

Bypass-only controlled sequence: USB absent → STANDALONE POWER installed → observer verified → relay off → TX disabled. All sources off before RJ45, harness, or path changes. Power removed before harness removal. Restore both independent +8V paths; voltage drop only after direct-path restoration; thermal only after direct-path restoration. Current no-control diagnostic cannot pass relay exercise gate and cannot pass UART exercise gate. No treadmill transfer and no MCU TX.

STOP gate: Any map, identity, current, drop, receive-only UART, bypass, thermal, or reset check fails. Power off; do not transfer the relay or transmit.

PASS gate: All eight paths, NC bypass, current, restored-path drop, receive-only observation, and thermal evidence pass. PASS — continue only to documentation closeout, never to treadmill control. **PASS — continue**

Operator: ______________________ Date: ______________________ Signed PASS: ______________________

Bypass sequence record: power removed before removal of the harness. Restore both independent +8V paths. Voltage drop only after direct-path restoration. Thermal only after direct-path restoration. The current no-control diagnostic cannot pass the relay exercise gate. The current no-control diagnostic cannot pass the UART exercise gate.